Name: Toppiiz ID: _____ No: _____

2110201 Quiz 1: Concept

Short Answers (1 point each)

T is represented by $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

$\xrightarrow[-P_1+P_3]{-P_2+P_1} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

$\xrightarrow[P_2+P_3]{} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

has pivot every column : 1-1

has pivot every row : onto

1. Consider a mapping $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x+y \\ y+z \\ z+x \end{bmatrix}$.

TRUE a. Is it true that T is one-to-one?

TRUE b. Is it true that T is onto?

2. Let a mapping T be described by a matrix A. Let A have 3 rows and 4 columns and that A is in a reduced row echelon form. Also, A does not have a row with all zeroes. ∴ A has 3 pivots, 1 free

FALSE a. Is it true that A represents a one-to-one mapping?

A is onto! but NOT 1-1

TRUE b. Is it true that A represents an onto mapping?

$\boxed{A \text{ is 1-1} \leftrightarrow A^T \text{ is onto}}$

TRUE c. Is it true that A^T represents a one-to-one mapping?

FALSE d. Is it true that A^T represents an onto mapping?

3. Let T be a mapping from $\mathbb{R}^3$ to $\mathbb{R}^2$. Let $S = \left\{\begin{bmatrix} 1 \\ 2 \end{bmatrix} a \,\middle|\, a \epsilon \mathbb{R}\right\}$. We know that the pre-image of S is $\mathbb{R}^3$. Write a matrix that represents T. (If there are multiple answers, you can write any correct one.)
(We know that the pre-image of S ***under the mapping T*** *is $\mathbb{R}^3$)* Let a matrix A represent T.

S = range of mapping T = set of all linear combination of columns of A.

∴ Columns of A must be multiple of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ → $A_{2\times3} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix}$ ※ (or $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix}$ etc.)

4. Consider the transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ where $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} a - 2y \\ b\left(2x - \frac{1}{4}\right) \end{bmatrix}$. If T is linear, what is the value of a and b.

T is linear, then

$T\left(2\begin{bmatrix} x \\ y \end{bmatrix}\right) = 2T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$

$\begin{bmatrix} a - 2(2y) \\ 2b(2x) - \frac{b}{4} \end{bmatrix} = 2\begin{bmatrix} a - 2y \\ 2bx - \frac{b}{4} \end{bmatrix}$

$\begin{bmatrix} a - 4y \\ 4bx - \frac{b}{4} \end{bmatrix} = \begin{bmatrix} 2a - 4y \\ 4bx - \frac{b}{2} \end{bmatrix}$

$\boxed{\begin{matrix} a = 2a \\ -\frac{b}{4} = \frac{b}{2} \end{matrix}}$ ∴ $\boxed{\begin{matrix} a = 0 \\ b = 0 \end{matrix}}$ ※

5. Three truck drivers went into a roadside café. One truck driver purchased four sandwiches, a cup of coffee, and ten doughnuts for \$8.45. Another driver purchased three sandwiches, a cup of coffee, and seven doughnuts for \$6.30. What did the third truck driver pay for a sandwich, a cup of coffee, and a doughnut?

S = price of a sandwich
C = price of a cup of coffee
D = price of a doughnut

$4S + C + 10D = 8.45$
$3S + C + 7D = 6.30$
$S + C + D = x$
find x!

$\begin{array}{c} \begin{array}{ccc} C & S & D \end{array} \\ \left[\begin{array}{ccc|c} 1 & 4 & 10 & 8.45 \\ 1 & 3 & 7 & 6.30 \\ 1 & 1 & 1 & x \end{array}\right] \end{array}$

$\xrightarrow[-P_1+P_3]{-P_1+P_2} \left[\begin{array}{ccc|c} 1 & 4 & 10 & 8.45 \\ 0 & -1 & -3 & -2.15 \\ 0 & -3 & -9 & x-8.45 \end{array}\right]$

$\xrightarrow[-3P_2+P_3]{} \left[\begin{array}{ccc|c} 1 & 4 & 10 & 8.45 \\ 0 & -1 & -3 & -2.15 \\ 0 & 0 & 0 & x-2 \end{array}\right]$

∴ $x - 2 = 0$
$x = 2$

$\boxed{\text{The 3rd driver paid \$2.00}}$ ※

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Show how to solve (2.5 points each)

1. Find the reduced row echelon matrix with the following homogeneous solution set:

$$\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}k+\begin{bmatrix}1\\5\\5\end{bmatrix}m \,\middle|\, k,m\in\mathbb{R}\right\}$$

2 Free variables

∴ 1 pivot variable

Only need 1 equation

↳ A, its row is perpendicular to the homogeneous solution.

Let $\begin{bmatrix}a\\b\\c\end{bmatrix}$ be a row of A, $\begin{bmatrix}a\\b\\c\end{bmatrix}\cdot\begin{bmatrix}1\\1\\1\end{bmatrix}=0$ and $\begin{bmatrix}a\\b\\c\end{bmatrix}\cdot\begin{bmatrix}1\\5\\5\end{bmatrix}=0$

$\boxed{\begin{aligned}a+b+c&=0\\a+5b+5c&=0\end{aligned}}$ Solve for a, b, c

∴ $\begin{bmatrix}a\\b\\c\end{bmatrix}\in\left\{\begin{bmatrix}0\\-1\\1\end{bmatrix}c \,\middle|\, c\in\mathbb{R}\right\}$

Let $c=-1$;

$$\begin{array}{c} \begin{array}{ccc} a & b & c \end{array} \\ \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 5 & 5 & 0 \end{array}\right] \end{array} \xrightarrow[-\rho_1+\rho_2]{} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 4 & 4 & 0 \end{array}\right] \xrightarrow[\frac{1}{4}\rho_2]{-\frac{1}{4}\rho_2+\rho_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

$\boxed{A=\begin{bmatrix}0 & 1 & -1\end{bmatrix}}$ ※

2. Given any three scalars a, b, and c forming the matrix $\begin{bmatrix}0 & c & -b\\-c & 0 & a\\b & -a & 0\end{bmatrix}$. Show that

   a. The linear transformation represented by that matrix is never one-to-one.

   b. The linear transformation represented by that matrix is never onto.

$$\begin{bmatrix}0 & c & -b\\-c & 0 & a\\b & -a & 0\end{bmatrix} \xrightarrow[\substack{b\rho_2\\c\rho_3}]{a\rho_1} \begin{bmatrix}0 & ac & -ab\\-bc & 0 & ab\\bc & -ac & 0\end{bmatrix}$$

$$\xrightarrow{\rho_2+\rho_3} \begin{bmatrix}0 & ac & -ab\\-bc & 0 & ab\\0 & -ac & ab\end{bmatrix}$$

$$\xrightarrow{\rho_1+\rho_3} \begin{bmatrix}0 & ac & -ab\\-bc & 0 & ab\\0 & 0 & 0\end{bmatrix}$$

ⓑ

$\boxed{\text{It has a zero-row, it's NOT onto.}}$ ※

ⓐ Case $abc=0$:

(either a or b or c is 0)

then it has free column(s).

Case $abc\neq 0$:

$$\xrightarrow{\rho_1\leftrightarrow\rho_2} \begin{bmatrix}-bc & 0 & ab\\0 & ac & -ab\\0 & 0 & 0\end{bmatrix}$$

$$\xrightarrow[\frac{1}{ac}\rho_2]{-\frac{1}{bc}\rho_1} \begin{bmatrix}1 & 0 & -\frac{a}{c}\\0 & 1 & -\frac{b}{c}\\0 & 0 & 0\end{bmatrix}$$

It has 1 free column.

∴ $\boxed{\text{It's NOT one-to-one.}}$ ※

OR $\begin{vmatrix}0 & c & -b\\-c & 0 & a\\b & -a & 0\end{vmatrix} = abc-abc=0$

Its determinant is zero, RRE $\neq I$

∴ RRE has $\underbrace{\text{free column(s)}}_{\text{ⓐ NOT 1-1 ※}}$ and $\underbrace{\text{zero-row(s)}}_{\text{ⓑ NOT onto ※}}$.